Abstract
PARTICIPANT: Kopal Tare
RESEARCH ADVISORS: Junming “Seraphina” Shi, Mohammad Shahrokh Esfahani
TITLE: REFRAME: Reference-Genome Reassessment for Accurate Mutation Evaluation

Mapping sequencing reads to a reference genome and calling variants are essential preprocessing steps in genomic and epigenomic analysis. However, multiple human references exist, and no standardized framework determines which is most appropriate. Because variant calls represent sample-reference differences, variant detection is inherently reference-dependent. Differences in completeness, ancestry representation, and structure can introduce bias and influence mapping accuracy, variant detection, and interpretation.

This project evaluates how reference choice affects whole-genome sequencing alignment and variant detection across three linear references: hg19, hg38, and T2T-CHM13. hg38 is an updated version of hg19, whereas T2T-CHM13 is a telomere-to-telomere reference that resolves previously missing or ambiguous regions. The Human Pangenome, a graph-based reference constructed from diverse haplotypes, was excluded because it requires a different alignment framework and is less interchangeable with linear references.

Three related whole-genome sequencing samples were analyzed as a family dataset, providing a shared genetic baseline for evaluating reference-dependent differences while reducing unrelated inter-individual variation. In parallel, 39 sarcoma samples were processed to determine whether these differences persisted in a cancer-relevant cohort.

Paired-end reads were aligned to hg19, hg38, and T2T-CHM13 using BWA-MEM. BAM files were processed with samtools, variants called with DeepVariant, and calls filtered with bcftools to retain high-quality calls (depth $\geq$10x). To standardize coordinates, hg19 and hg38 VCF files were lifted over to CHM13v2 using Picard LiftoverVcf, enabling identification of shared and reference-specific variants.

Across datasets, mapping rates exceeded 99.7%, but resulting variant calls differed among references. hg19 and hg38 showed the strongest concordance, whereas T2T-CHM13 produced distinct profiles, including allele frequency shifts, heterozygous-to-homozygous ratios, and chromosome-specific distributions. More than 70% of variants were shared, but reference-specific differences persisted, especially in complex regions.

These findings demonstrate that reference choice influences variant detection and interpretation and should be carefully considered when developing, validating, and benchmarking genomic analysis pipelines.